

CS 613: NLP - Assignment 2

Tokenizer & Model Training

Total Marks: 100 Pts
November 17, 2024

Problem Statement

Task 1: Tokenizer Training

1.1 Tokenizer Training

Dataset Samples Description: The curated scrapped data by all the team members was used to train the tokenizer. The dataset was called `all_final_dataset`. A random sampling from this dataset (with a fixed seed) was used to make datasets of fixed sizes. Additionally, a dataset containing only news articles was compiled called `all_news_dataset`, along with standalone datasets like Sangrah and Nepali Deva, as they were substantial in size on their own.

- Dataset Sample 1: `all_final_dataset` (200 MB)
- Dataset Sample 2: `all_final_dataset` (350 MB)
- Dataset Sample 3: `all_final_dataset` (500 MB)
- Dataset Sample 4: `all_final_dataset` (750 MB)
- Dataset Sample 5: `all_final_dataset` (1 GB)
- Dataset Sample 6: `all_news_scrapped` (350 MB)
- Dataset Sample 7: `Sangrah` (750 MB)
- Dataset Sample 8: `Nepali_Deva` (1.2 GB)

Tokenizer Details:

Tokenizer	Algorithm	Vocab Size	Dataset Information
Tokenizer 1	sentencePiece	49,152	<code>all_final_dataset</code> (200 MB)
Tokenizer 2	sentencePiece	49,152	<code>all_final_dataset</code> (350 MB)
Tokenizer 3	sentencePiece	49,152	<code>all_final_dataset</code> (500 MB)
Tokenizer 4	sentencePiece	49,152	<code>all_final_dataset</code> (750 MB)
Tokenizer 5	sentencePiece	49,152	<code>all_final_dataset</code> (1 GB)
Tokenizer 6	sentencePiece	49,152	<code>all_news_scrapped</code> (350 MB)
Tokenizer 7	sentencePiece	50,000	<code>Sangrah</code> (750 MB)

1.2 Fertility Score Calculation

Methodology: Fertility scores were calculated for the trained tokenizers on both their respective training datasets and a held-out dataset (`all_news_scrapped`).

Results when tested on the same dataset it was trained on:

Tokenizer	Fertility Score	Dataset Info (Train)	Dataset Info (Test)
Tokenizer 1	1.2911565101	all_final_dataset (200 MB)	all_final_dataset (200 MB)
Tokenizer 2	1.2952214366	all_final_dataset (350 MB)	all_final_dataset (350 MB)
Tokenizer 3	1.2952510265	all_final_dataset (500 MB)	all_final_dataset (500 MB)
Tokenizer 4	1.2964268691	all_final_dataset (750 MB)	all_final_dataset (750 MB)
Tokenizer 5	1.2960094995	all_final_dataset (1 GB)	all_final_dataset (1 GB)
Tokenizer 7	1.2469413133	Sangrah (200 MB)	Sangrah (200 MB)

Results when tested on a fixed held-out dataset (all_news_scrapped):

Tokenizer	Fertility Score	Dataset Info (Train)	Dataset Info (Test)
Tokenizer 1	1.2529094399	all_final_dataset (200 MB)	all_news_scrapped (350 MB)
Tokenizer 2	1.2515642463	all_final_dataset (350 MB)	all_news_scrapped (350 MB)
Tokenizer 3	1.2509406335	all_final_dataset (500 MB)	all_news_scrapped (350 MB)
Tokenizer 4	1.2504782866	all_final_dataset (750 MB)	all_news_scrapped (350 MB)
Tokenizer 5	1.2515644638	all_final_dataset (1 GB)	all_news_scrapped (350 MB)

Task 2: Model Training

2.1 Model Selection

Model Architecture Chosen: The architecture chosen is **LLaMAForCausalLM**, which is based on the transformer architecture optimized for causal language modeling tasks. It employs a multi-head self-attention mechanism with rotary embeddings, RMS normalization, and an efficient feed-forward network.

Adjustments Made: To reduce the total number of parameters to below 100M, the following modifications were made:

- Reduced the hidden_size to **512** (compared to higher defaults in larger models).
- Set number of attention heads to **8** with a head dimension of **64**.
- Kept **16 layers** in the model (num_hidden_layers).
- Set intermediate_size to **1024**, which balances the computational load in the feed-forward layers.
- Used a vocabulary size of **50,000**, optimized for the dataset size.
- Excluded additional biases in feed-forward layers (mlp_bias: false) to reduce parameters.

Total Parameters: The final model has approximately **86M parameters**, which is within the 100M constraint.

2.2 Tokenization of Dataset

Best Tokenizer Chosen: The tokenizer 7 (trained on Sangrah (750 MB)) was chosen for the model training.

Reason for Choice: It had a perplexity of around **1.2469413133**, which was the lowest among all the tokenizers trained on different datasets. Thus, we choose it as the best tokenizer for the model training.

2.3 Model Training

Training Details:

- Number of Epochs: **10**
- Batch Size: **8**

Perplexity Results per Epoch:

The following table shows the training loss, validation loss, and perplexity at various steps during model training. Every 500 steps correspond to 0.1 epoch.

Steps	Training Loss	Validation Loss	Perplexity
500	7.918729	7.703134	2102.312241
1000	7.529400	7.505957	1818.844430
1500	7.207300	7.229811	1437.476760
2000	6.884200	6.893766	986.108044
2500	6.688900	6.741419	838.426617
3000	6.493600	6.508142	670.579000
3500	6.371900	6.391015	591.100484
4000	6.250200	6.223870	504.652362
4500	6.111200	6.113407	451.556748
5000	5.971700	6.002983	404.633863
5500	5.877500	5.910859	366.748795
6000	5.783300	5.816735	335.873564
6500	5.687900	5.723983	310.632056
7000	5.630000	5.672997	290.905210
7500	5.577500	5.599084	274.989125
8000	5.524400	5.547181	256.513335
8500	5.469400	5.487976	241.756643
9000	5.444300	5.444356	231.448171
9500	5.385400	5.398842	222.632577
10000	5.327800	5.347515	210.085508
10500	5.276100	5.306705	202.228561
11000	5.227900	5.265094	193.464456
11500	5.157300	5.214261	188.233344
12000	5.184900	5.194471	180.272732
12500	5.113100	5.126363	173.604526
13000	5.041700	5.132655	169.466479
13500	5.028200	5.092561	163.415618
14000	5.021400	5.080809	160.904193
14500	5.015000	5.068379	157.665767
15000	5.016100	5.031531	153.167311
15500	4.993000	5.009063	149.152026
16000	4.924900	4.986255	146.387131
16500	4.890000	4.967847	142.557062
17000	4.855600	4.947319	140.796985
17500	4.850000	4.915397	137.633544
18000	4.834600	4.908449	135.429160
18500	4.832000	4.885120	133.049802
19000	4.840000	4.873579	130.788174
19500	4.774500	4.853282	128.125492
20000	4.688000	4.849658	127.696651

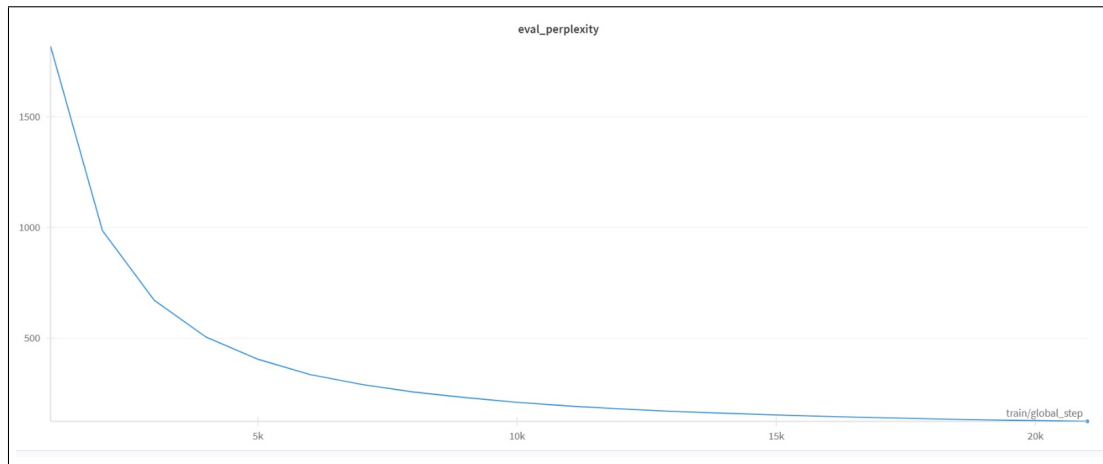


Figure 1: Eval Perplexity v/s Training Step

The above graph shows how evaluation perplexity keeps decreasing as the training progresses.

2.4 Model Testing

Testing Prompts & Outputs:

As $\text{\LaTeX}$ does not support Devnagri characters, all the prompts and generated texts are documented in the Google Doc file linked here: **Link to Google Doc**

2.5 Other Trained Models

The best-trained model was chosen for the final submission. Below are the details of the other models that were trained:

- **Nepali LLM Version 1:**

- **Trained on:** `all_final_dataset` (500 MB)
- **Model Architecture:** LlamaForCausalLM
- **Parameters:** 86.000128 Million
- **Tokenizer Used:** Tokenizer 4 (trained on `all_final_dataset` (750 MB))
- **Perplexity:** 4000
- **Epochs:** 10
- **Training Split:** 80:20

- **Nepali LLM Version 2:**

- **Trained on:** `all_final_dataset` (200 MB)
- **Model Architecture:** LlamaForCausalLM
- **Parameters:** 86.000128 Million
- **Tokenizer Used:** Tokenizer 4 (trained on `all_final_dataset` (750 MB))
- **Perplexity:** 171
- **Epochs:** 10
- **Training Split:** 90:10

References

- Aamod Thakur GitHub Repository

Code Link

- **GitHub Repository:** [GitHub Link](#)

Work Distribution

- **Aditya Mehta (Roll Number: 22110017)**
 - Tokenized the `all_news_scrapped` dataset and identified duplicates with other datasets.
 - Worked on the tokenization process of the `Nepali_Deva` text corpus.
 - Prepared the entire report.
- **Daksh Jain (Roll Number: 22110066)**
 - Tokenized a Nepali Wikipedia-scraped dataset and evaluated fertility scores to determine the best tokenization strategy.
 - Tested scalability by tokenizing datasets of various sizes.
 - Ran prompt tests on the final model to validate its performance.
- **Hriday Ruparel (Roll Number: 22110099)**
 - Wrote scripts for dataset merging, tokenization, and model training.
 - Trained tokenizers on the scraped dataset.
 - Trained an 80M model on the scraped dataset.
- **Kishan Ved (Roll Number: 22110122)**
 - Trained five different tokenizers and identified the one with the best fertility score.
 - Trained an 86M parameter model on the scraped dataset.
 - Trained a tokenizer on the Sangrah dataset and tokenized it.
 - Trained an 86M parameter model on the Sangrah dataset.
- **Summet Sawale (Roll Number: 22110234)**
 - Assisted in writing code for tokenizing text into tensors of context length.
 - Tested models using prompts.
 - Documented testing results.
 - Downloaded the scraped data from the server.